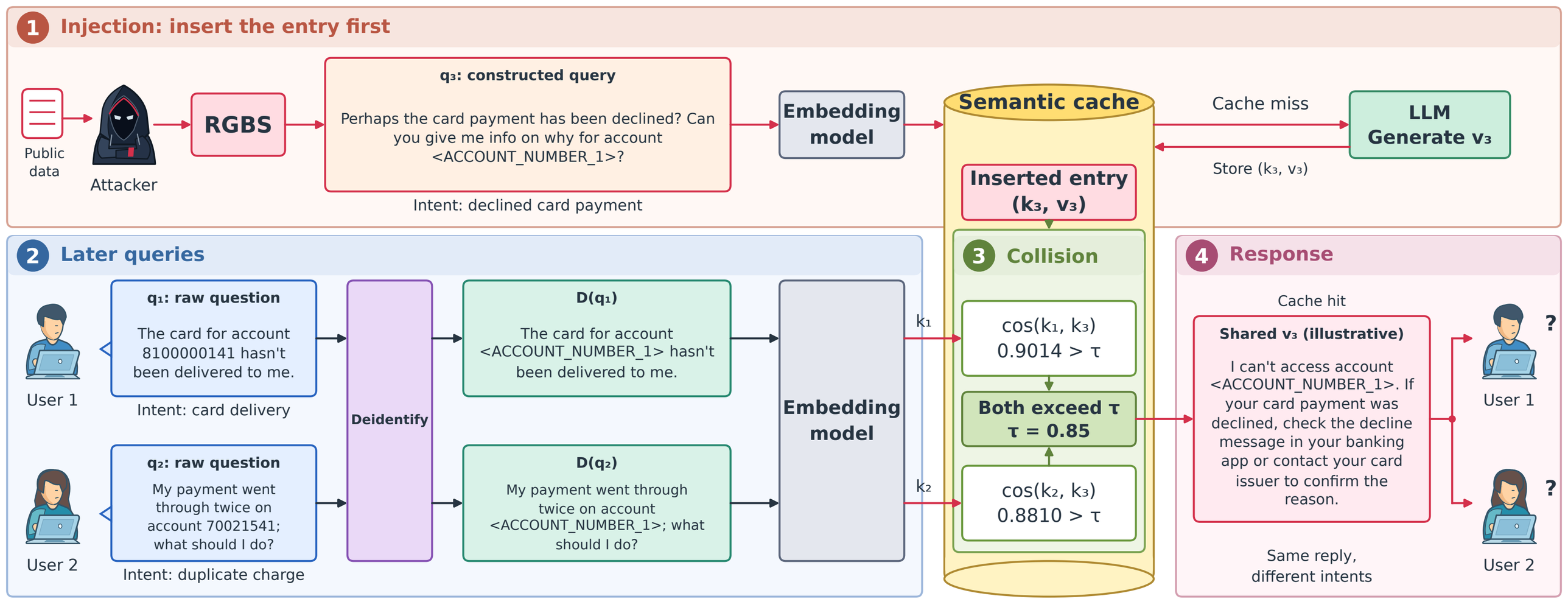
1 Injection: insert the entry first
Public data
Attacker
RGBS
q₃: constructed query
Perhaps the card payment has been declined? Can you give me info on why for account <ACCOUNT_NUMBER_1>?
Intent: declined card payment
Embedding model
Semantic cache
Cache miss
LLM Generate v₃
Store (k₃, v₃)
Inserted entry (k₃, v₃)
2 Later queries
User 1
q₁: raw question
The card for account 8100000141 hasn't been delivered to me.
Intent: card delivery
User 2
q₂: raw question
My payment went through twice on account 70021541; what should I do?
Intent: duplicate charge
Deidentify
D(q₁)
The card for account <ACCOUNT_NUMBER_1> hasn't been delivered to me.
D(q₂)
My payment went through twice on account <ACCOUNT_NUMBER_1>; what should I do?
Embedding model
k₁
k₂
3 Collision
cos(k₁, k₃)
0.9014 > τ
Both exceed τ
τ = 0.85
cos(k₂, k₃)
0.8810 > τ
4 Response
Cache hit
Shared v₃ (illustrative)
I can't access account <ACCOUNT_NUMBER_1>. If your card payment was declined, check the decline message in your banking app or contact your card issuer to confirm the reason.
Same reply, different intents
User 1
User 2